DAURES GREEN HYDROGEN VILLAGE

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With funding from the:
Federal Ministry of Research, Technology and Space

SAF Production Pathways from Namibian Encroacher Bush

Three thermochemical pathways can convert solid biomass like encroacher bush into Sustainable Aviation Fuel (SAF). Each pathway below summarises the process chemistry, operating parameters and co-product handling.

1. FISCHER-TROPSCH (FT) SYNTHESIS

Process Overview

- Biomass combustion — Encroacher bush Otjikoto 40 MW_{el} power plant; CO_2-rich flue gas produced along electricity.
- Post-combustion carbon capture (MEA solvent); compressed and liquefied for road transport to eFuels facility.
- Reverse Water-Gas Shift (RWGS) reaction — $CO_2 + H_2$ over Cu or Fe/Al_2O_3 catalysts, yields syngas with $H_2/CO \approx 2:1$.

$CO_2 + H_2 \rightleftharpoons CO + H_2O$ ($\Delta H = +42.1$ kJ/mol) [1]

- Low-temperature FT synthesis — for long-chain paraffins production through polymerisation. Low aromatics and S-content, ideal SAF precursor [2].
- Hydrocracking & fractionation — Syncrude upgraded to ASTM D7566 SAF, diesel, and gasoline fractions.

Combustion air excess	20%
CO_2 capture efficiency	85-95% (MEA)
Flue gas CO_2 conc.	13–15 vol%
RWGS temperature	400 - 700 °C
FT temperature	200 - 240 °C (LTFT)
FT catalyst	Cobalt-based
Feedstock intensity	3.10 kg dry bush / kg SAF
H_2 intensity	0,55 kg H2 / kg eFuel

Table 1: FT Operating Parameters

Advantages

- High carbon efficiency
- Potential for negative carbon emissions
- Electricity co-production
- Mature technology
- Biogenic CO_2 reuse (BECCU)

Challenges

- High water demand
- High capital cost
- CO_2 transport logistics

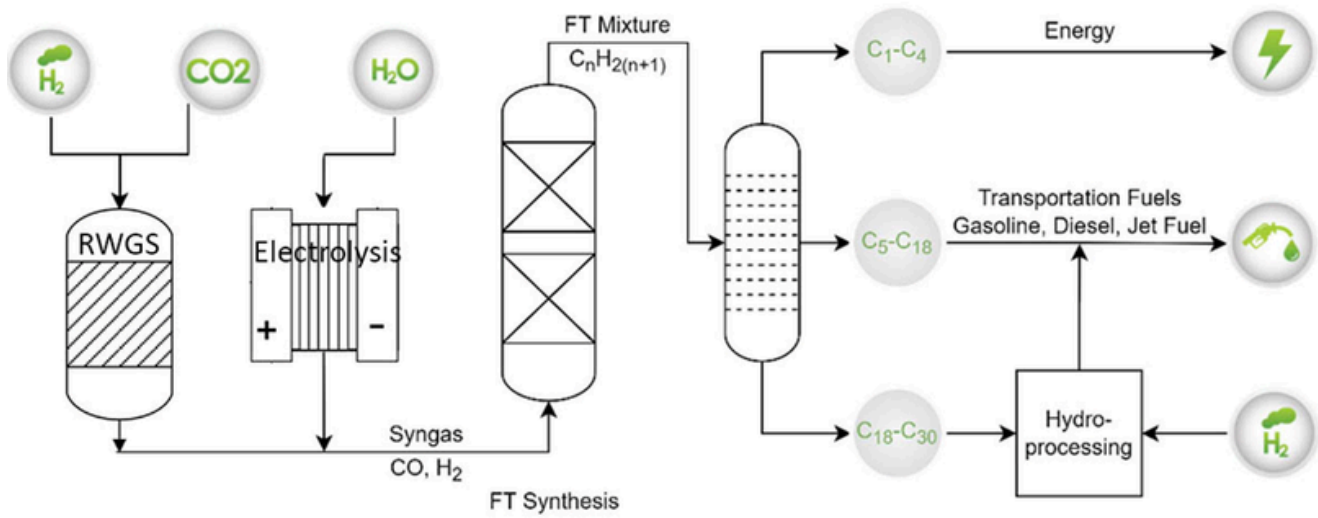

Figure. 1: Schematic flow diagram of the FT synthesis pathway,. Adapted from ANL [3]

2. HYDROTHERMAL LIQUEFACTION (HTL)

Process Overview

- Slurry preparation — Biomass chipped and mixed with water forming solid slurry. No drying required.
- HTL — Biomass converted in sub-/ supercritical water (acting as solvent + reactant) via depolymerization, hydrolysis, decarboxylation, and recombination reactions to yield biocrude. [4]
- Phase separation — Multiphase output separated into bio-crude (19–40 wt%), hydrochar (4–39 wt%), aqueous phase, and process gas (mainly CO_2). [5]
- Anaerobic digestion (AD) — Aqueous phase treated by AD produces biogas, supplying internal heat to the process.
- Hydrodeoxygenation (HDO) & upgrading — Bio-crude hydrotreated (NiMo/CoMo catalysts, high-pressure H_2) to remove the high oxygen content, then cracked and fractionated to SAF.

Temperature	250–370 °C
Pressure	5–22 MPa
Residence time	15–60 minutes
RWGS temperature	400–700 °C
Slurry solids	10–20 wt%
Bio-crude HHV	28–38 MJ/kg
Bio-crude O content	22–31 wt% (woody biomass)
Feedstock intensity	11.77 kg dry bush / kg SAF
H_2 intensity	0,07 kg H2/kg eFuel

Table 2: HTL Operating Parameters

Advantages

- Tolerates feedstock variability
- Nutrient recovery possible
- Internal heat via AD biogas
- Hydrochar coproduct valuable in agricultural and cement industry.

Challenges

- High feedstock intensity
- High equipment cost (pressure)
- Wastewater management
- Dry biomass requires added water

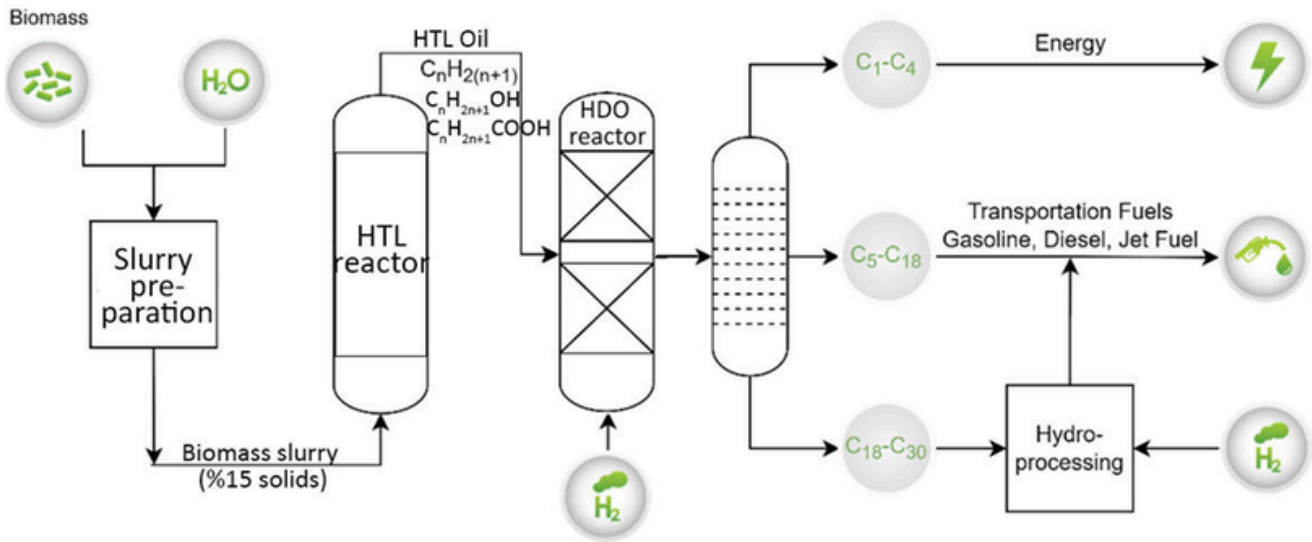

Figure. 2: Process flow diagram of HTL. Adapted from ANL [3]

3. FAST PYROLYSIS (FP)

Process Overview

- Pre-treatment — Biomass chipped and mechanically ground. Low moisture (8.5 wt%) bypasses need for pre-drying.
- Fast pyrolysis reactor — Rapid thermal decomposition in absence of oxygen. Yields: 60-70% oil, 15-25% char, 10-20% non-condensable gases. [6]
- Internal heat integration — Char and ash combustion provides reactor heat. Potential electricity generation with surplus off-gas.
- Oil quenching & stabilisation — Vapours rapidly condensed to produce the oil.
- HDO upgrading — Oil hydrotreated with green H_2 (Ru/C or NiMo catalysts) to remove oxygen. Hydrocracked and fractionated to SAF.

Temperature	400–600 °C
Heating rate	>1,000 °C/s (up to 10,000)
Vapour residence time	<2 seconds
Bio-oil HHV	16–19 MJ/kg
Bio-oil O content	35–40 wt%
Feedstock intensity	7.51 kg dry bush / kg SAF
H_2 intensity	0,1 kg H2/kg eFuel

Table 3: FP Operating Parameters

Advantages

- Internal energy self-sufficiency
- Decentralised plant potential
- No pre-drying required for dry biomass

Challenges

- Corrosive, unstable oil
- Higher HDO hydrogen demand than HTL
- High biogenic process losses
- Upgrading costs significant

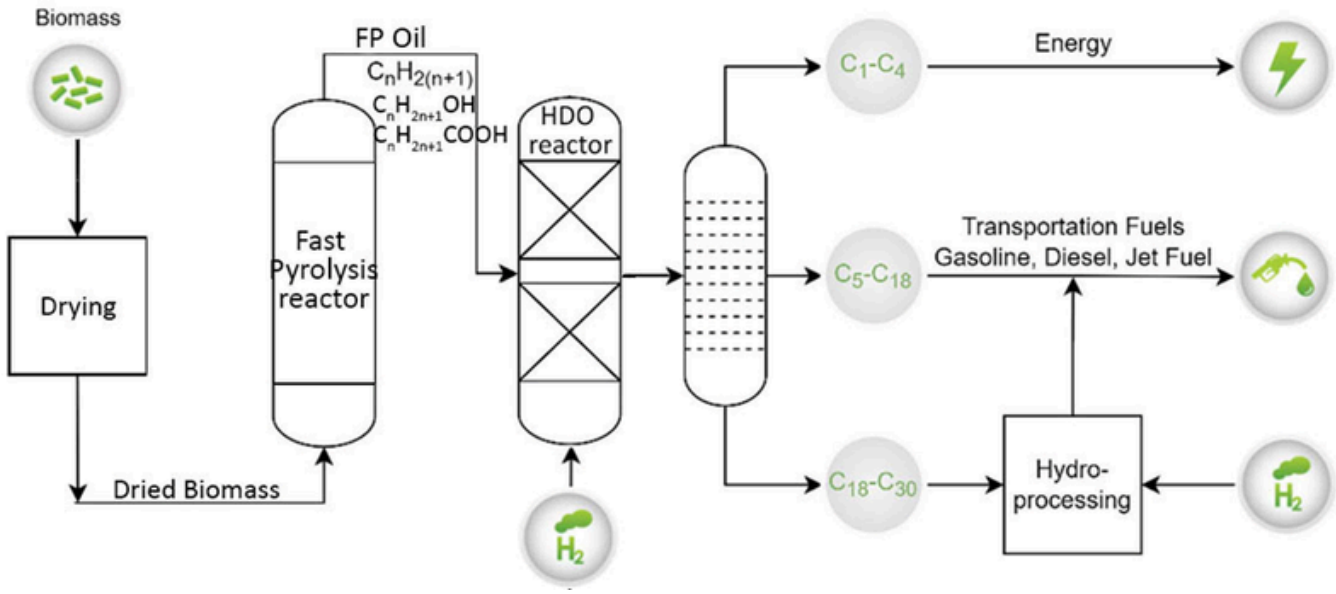

Figure. 3: Flow diagram of fast pyrolysis. Adapted from ANL [3]

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https://github.com/IER-Hy4Daures/Fact-Sheets